Chapter 1

Units and Measurements

■ Key Points

1. Physical quantities = measurable quantities (length, mass, time, etc.)

2. SI Units: Length=metre(m), Mass=kilogram(kg), Time=second(s), Current=ampere(A), Temp=kelvin(K), Amount=mole(mol), Luminosity=candela(cd)

3. Supplementary: Plane angle=radian(rad), Solid angle=steradian(sr)

4. Dimensional Formula: Expresses a physical quantity in terms of M, L, T

5. Uses of Dimensions: Check correctness of equations, derive formulas, convert units

6. Significant Figures: All certain digits + one uncertain digit in a measurement

7. Rules: All non-zero digits are significant; zeros between non-zero digits are significant

8. Errors: Systematic error (constant), Random error (fluctuating), Gross error (human mistake)

9. Absolute Error = |True value - Measured value|

10. Relative Error = Absolute Error / True Value

11. Percentage Error = Relative Error × 100%

12. For addition/subtraction: Add absolute errors

13. For multiplication/division: Add relative errors

■ Important Formulas

Formula	Description
$[v] = [M\blacksquare L^{1}T\blacksquare^{1}]$	Velocity
$[a] = [M\blacksquare L^{1}T\blacksquare^{2}]$	Acceleration
$[F] = [M^{1}L^{1}T\blacksquare^{2}]$	Force
$[W] = [M^{1}L^{2}T\blacksquare^{2}]$	Work/Energy
$[P] = [M^{1}L^{2}T\blacksquare^{3}]$	Power